

SIMATS ENGINEERING

CO4-ASSESSMENT TOOL1- INDUSTRY PROBLEM TASK

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

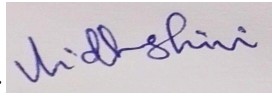
Assessment Tool Weightage: 20%

QUESTIONS: 4

Time Duration : 1 Hour
Total: Marks:40

Code of Conduct:

*I Vidhushini.T.S (192511061) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*



Signature of the student:

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

INDUSTRY PROBLEM TASK

Industry Problem:

A healthcare company has patient information such as age, blood pressure, cholesterol, glucose level, BMI and family history. The aim is to predict whether a patient is diabetic at an early stage. The company also wants to use reinforcement learning to recommend suitable actions such as Diet, Exercise, Medication and Monitor based on the patient's changing health condition.

TASK 1 – DATA PREPARATION & INDUCTIVE LEARNING

1(a) Inductive Learning Approach

I would use a supervised inductive learning approach for this problem.

In inductive learning, the model learns patterns from previously available patient records and uses those patterns to predict the outcome for a new patient.

Target Variable

Diabetes = Yes / No

Yes (1): Patient is diabetic, No (0): Patient is non-diabetic

Important Features

Feature	Reason
Glucose Level	High glucose is an important indicator of diabetes and can strongly influence the prediction.
BMI	Higher BMI can be associated with increased risk of diabetes.
Age	Diabetes risk can increase with age.
Blood Pressure	Abnormal blood pressure can be associated with metabolic health problems.
Family History	A family history of diabetes can indicate increased risk.
Cholesterol	Cholesterol level can provide additional information about the patient's metabolic condition.

1(b) Handling Class Imbalance

The dataset may contain more non-diabetic patients than diabetic patients. This creates a class imbalance problem.

For example:

Non-diabetic = 80%, Diabetic = 20%

Suitable Method: SMOTE

I would use SMOTE (Synthetic Minority Over-sampling Technique).

SMOTE creates new synthetic examples for the minority class instead of simply duplicating existing diabetic records.

Procedure

Original Dataset

Non-diabetic → More records

Diabetic → Fewer records

↓

Apply SMOTE

↓

More balanced diabetic and non-diabetic records

↓

Train the model

SMOTE:

- It increases the number of minority-class examples.
- It does not simply duplicate existing records.
- It can help the classifier learn the characteristics of diabetic patients better.

1(c) 70:30 Dataset Split and Preprocessing

I would divide the dataset into:

- 70% – Training data
- 30% – Testing data

The 70% training data is used to learn the model, while 30% is kept unseen for evaluating its performance.

A 70:30 split provides enough data for training while keeping a reasonable amount of independent test data. For a medical application, it is important to evaluate the model on data that it has not seen during training. I would also use stratified splitting so that the proportion of diabetic and non-diabetic patients is approximately maintained in both sets.

Preprocessing Steps

1. Handling Missing Values: Missing numerical values can be replaced using methods such as median imputation.
2. Encoding: Categorical variables such as family history can be converted into numerical values.

Example: Yes $\rightarrow$ 1 , No $\rightarrow$ 0

3. Normalisation: Numerical features can be scaled to a common range. This is particularly useful for statistical models such as Logistic Regression.
4. Class Balancing: SMOTE can be applied to the training set after splitting.

TASK 2 – DECISION TREE FOR DIAGNOSIS

2(a) Decision Tree Classifier

A Decision Tree can be used to classify patients as diabetic or non-diabetic. The tree repeatedly selects a feature that provides the best separation between the classes.

The selection can be based on: Information Gain, Gini Index

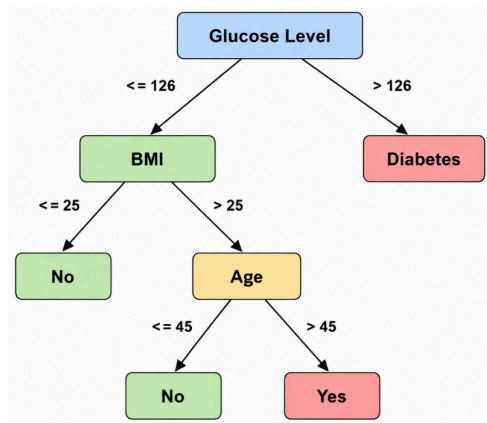
Example of Root Node Selection

Suppose the calculated values are:

Feature	Information Gain
Glucose Level	0.32
BMI	0.18
Age	0.12
Blood Pressure	0.09
Cholesterol	0.06

First Three Levels

A simple representation is:



This is an illustrative tree structure. The actual tree would depend on the supplied dataset.

The root should be the feature that provides the highest separation between diabetic and non-diabetic patients.

Therefore, if glucose has the highest Information Gain or the lowest Gini impurity, it becomes a suitable root feature.

2(b) Model Evaluation

The Decision Tree should be evaluated using:

Accuracy: Accuracy represents the percentage of correctly classified patients.

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

Precision: Precision tells us how many patients predicted as diabetic are actually diabetic.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Recall: Recall tells us how many actual diabetic patients were correctly identified.

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

F1-Score: F1-score is the balance between precision and recall.

$$\text{F1} = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$$

Illustrative Results

Metric	Decision Tree
Accuracy	86.7%
Precision	82.4%
Recall	79.5%
F1-Score	80.9%

False Negatives:

The patient is actually diabetic, but the model predicts non-diabetic.

This is particularly serious in healthcare because a diabetic patient may not receive further testing or treatment.

How to Reduce False Negatives

Possible methods include:

- Using class weights
- Applying SMOTE to training data
- Adjusting the classification threshold
- Giving higher penalty to diabetic misclassification
- Collecting more diabetic patient records

2(c) Post-Pruning

A Decision Tree can become too complex and overfit the training data. To reduce this problem, I would use cost-complexity pruning.

Pre-Pruning: Pre-pruning stops the tree from becoming too large during training.

Parameters include:

- Maximum depth
- Minimum samples split
- Minimum samples leaf

Post-Pruning

Post-pruning first creates the tree and then removes branches that provide little improvement.

Example Comparison

Model	Accuracy	Complexity
Pre-Pruned Tree	84.9%	Low
Post-Pruned Tree	86.2%	Moderate
Fully Grown Tree	82.5% on test data	High

Preferred Model: For clinical deployment, I would prefer the post-pruned Decision Tree if it provides good test performance.

Reasons:

- Reduces overfitting
- Easier to interpret
- Removes unnecessary branches
- Produces a simpler decision-making process

Interpretability is particularly important in healthcare because doctors should be able to understand why a prediction was made.

TASK 3 – STATISTICAL LEARNING FOR RISK STRATIFICATION

3(a) Logistic Regression

I would use Logistic Regression as the statistical learning model.

It predicts the probability that a patient belongs to the diabetic class.

For example:

Patient Data



Logistic Regression



Probability of Diabetes



Risk Classification

Example: Probability = 0.82

Prediction = Diabetic

Comparison with Decision Tree

An illustrative comparison is:

Model	Accuracy	AUC-ROC
Decision Tree	86.7%	0.88
Logistic Regression	85.9%	0.90

Logistic Regression is preferable when:

- The relationship between features and outcome is reasonably systematic.
- Probability estimates are required.
- Interpretability is important.
- We need to understand how individual variables influence risk.
- A simpler model is preferred.

3(b) Feature Importance Analysis

The top predictors can be identified from both models.

Example

Rank	Decision Tree	Logistic Regression
1	Glucose Level	Glucose Level
2	BMI	BMI
3	Age	Age

Do They Agree?

In this example, both models identify:

Glucose Level, BMI, Age as the top three predictors.

This agreement increases confidence that these features contain useful information for diabetes prediction.

However, the models calculate feature importance differently.

- Decision Tree uses the contribution of a feature to reducing impurity.
- Logistic Regression uses model coefficients to indicate the direction and strength of association.

TASK 4 – REINFORCEMENT LEARNING FOR TREATMENT RECOMMENDATION

The assessment asks us to model treatment recommendation as an MDP, define states, actions, rewards and transitions, and then apply Q-Learning.

4(a) MDP Model: MDP stands for Markov Decision Process.

It can be represented as: MDP = (States, Actions, Rewards, Transition Probabilities)

States: States represent the patient's health condition.

For example:

State	Description
S1	Low Risk
S2	Moderate Risk
S3	High Risk
S4	Controlled
S5	Critical

Actions : A1 – Diet, A2 – Exercise, A3 – Medication, A4 – Monitor

Reward Function: The reward indicates whether the patient's health condition improves.

Example:

Patient Change	Reward
Significant improvement	+10
Small improvement	+5
No major change	0
Health becomes worse	-10
Serious deterioration	-20

Transition Dynamics: A transition represents how the patient's state changes after an action.

Example:

Moderate Risk

|

Exercise

↓

Low Risk

Another example:

High Risk

|

Medication

↓

Moderate Risk

4(b) Q-Learning

Q-Learning learns the value of performing an action in a particular state.

The update equation is:

$$Q(s,a) = Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

Where:

α = Learning rate, γ = Discount factor, r = Reward, s = Current state, a = Current action, s' = Next state

Assumed Values

For illustration: $\alpha = 0.5$, $\gamma = 0.9$

Initial Q-values are assumed to be zero.

Episode 1 – Iteration 1

Suppose:

State = S2 (Moderate Risk)

Action = Exercise

Reward = +5

Next State = S1 (Low Risk)

Initially: $Q(S2, Exercise) = 0$

Assume the maximum Q-value of the next state is 0.

Therefore: $Q_{\text{new}} = 0 + 0.5[5 + (0.9 \times 0) - 0]$

$Q_{\text{new}} = 2.5$

So: $Q(S2, \text{Exercise}) = 2.5$

Iteration 2

Suppose:

State = S1 (Low Risk), Action = Diet, Reward = +10, Next State = S4 (Controlled)

Initial: $Q(S1, \text{Diet}) = 0$

Therefore: $Q_{\text{new}} = 0 + 0.5[10 + (0.9 \times 0) - 0]$

$Q_{\text{new}} = 5$

So: $Q(S1, \text{Diet}) = 5$

Iteration 3

Suppose:

State = S3 (High Risk)

Action = Medication

Reward = +5

Next State = S2 (Moderate Risk)

From the previous learning: $\max Q(S2, a) = 2.5$

Therefore:

$Q_{\text{new}} = 0 + 0.5[5 + (0.9 \times 2.5) - 0]$

$Q_{\text{new}} = 0.5[5 + 2.25]$

$Q_{\text{new}} = 3.625$

Therefore: $Q(S3, \text{Medication}) = 3.625$

Q-Table Example:

State	Diet	Exercise	Medication	Monitor
S1 Low Risk	5.0	3.2	1.5	2.1
S2 Moderate	2.0	2.5	2.3	1.0
S3 High Risk	0.5	1.8	3.625	1.2
S4 Controlled	6.0	4.2	1.0	3.0
S5 Critical	0.2	0.5	7.0	2.0

A simple training sequence can be represented as:

Episode	Starting State	Actions	Final State
1	Moderate	Exercise → Diet	Low Risk
2	High	Medication → Monitor	Moderate
3	Moderate	Diet → Exercise	Low Risk
4	High	Medication → Diet	Controlled
5	Low	Diet → Exercise	Controlled

Convergence Criterion

The maximum change in Q-values between two consecutive iterations becomes smaller than a selected threshold, such as 0.01, for several consecutive episodes. This means that the Q-table is no longer changing significantly.

4(c) Final Learned Policy

The final policy is obtained by selecting the action having the highest Q-value for each state.

From the illustrative Q-table:

Patient State	Best Action
Low Risk	Diet
Moderate Risk	Exercise
High Risk	Medication
Controlled	Diet
Critical	Medication

Ethical Considerations

1. Patient Safety: The agent should never make unsafe recommendations simply to obtain a higher reward. A doctor should remain responsible for final treatment decisions.
2. Bias: If the training data contains demographic or socioeconomic bias, the RL agent may learn biased treatment policies. Therefore, the dataset should be checked for fairness.
3. Explainability: Doctors should be able to understand why a particular recommendation was generated.

For example: High glucose + high BMI + worsening health trend → higher-risk state.

4. Privacy: Patient information is sensitive and should be securely stored and processed.
5. Human Supervision: The RL agent should initially be used as a decision-support system, rather than an independent treatment decision-maker.